**SDS - SAFETY DATA SHEET -** In accordance with Globally Harmonized System of Classification and Labelling of Chemicals (GHS) – Chapter 1.5 and Annex 4

## 1 – IDENTIFICATION

**GHS Product identifier:** ICHIBAN

**Recommended use of the chemical and restrictions on use:** Fungicide for use in farming.

**Supplier's details:** Pilarquim (Shanghai) Co.,Ltd.

**Address**: 1500 Hang-Tang Road, Jin-Hui Town, Feng Xian District, Shanghai 201405 – CHINA

**Phone number(s):** (+86-21) 5758-9888

**Fax:** (+86-21) 5758-9666

**Emergency phone number:** (+86-21) 5758-9888

## 2 – HAZARD IDENTIFICATION

**Classification of the substance or mixture**:

Acute toxicity – Inhalation - Category 1

Serious eye damage/eye irritation – Category 2A

Carcinogenicity – Category 2*

Specific target organ toxicity – single exposure – Category 3

Hazardous to the aquatic environment – Acute – Category 1

Hazardous to the aquatic environment – Chronic – Category 1

* Classification due to the presence of Chlorothalonil. See section 8 and 11 for more information.

**Classification system adopted** Globally Harmonized System of Classification and Labelling of Chemicals (GHS), United Nations.

**GHS label elements, including precautionary statements**

**Pictograms**:

**Signal word**: DANGER

**Hazard statement(s)**:

H319 Causes serious eye irritation.

H330 Fatal if inhaled.

H335 May cause respiratory irritation.

H351 Suspected of causing cancer.

H410 Very toxic to aquatic life with long lasting effects.

**Precautionary statement(s)**:

**PREVENTION**:

P201 Obtain special instructions before use.

P202 Do not handle until all safety precautions have been read and understood.

P260 Do not breathe mist and vapors.

P264 Wash hands thoroughly after handling.

P271 Use only outdoors or in a well-ventilated area.

P273 Avoid release to the environment.

P280 Wear protective gloves, protective clothing, eye protection and face protection.

P284 [In case of inadequate ventilation] wear a respiratory protection.

**RESPONSE**:

P304 + P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P308 + P313 IF exposed or concerned: Get medical attention.

P310 Immediately call a POISON CENTER or a doctor.

P312 Call a POISON CENTER or a doctor if you feel unwell.

P320 Specific treatment is urgent.

P321 Specific treatment.

P337 + P313 If eye irritation persists: Get medical advice/attention.

P362 + P364 Take off contaminated clothing and wash it before reuse.

P391 Collect spillage.

**STORAGE**:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

P405 Store locked up.

**DISPOSAL**:

P501 Dispose of contents in accordance with local regulations.

## 3 – COMPOSITION/INFORMATION ON INGREDIENTS

**MIXTURES**

**Ingredients or impurities contributing to the hazard**:

| Components | Concentration | CAS number |
|---|---|---|
| Chlorothalonil | 72% | 1897-45-6 |

## 4 – FIRST-AID MEASURES

**Routes of exposure**

**Inhalation**: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician if you feel unwell. Take this SDS.

**Skin**: Wash exposed skin with enough water to remove the material. In case of irritation: Consult a doctor. Take this SDS.

**Eye**: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. If eye irritation persists: Get medical advice/attention. Take this SDS.

**Ingestion**: Do not induce vomiting. Never give anything by mouth to an unconscious person. Rinse the victim's mouth with water. Call a POISON CENTER or doctor/physician If you feel unwell. Take this SDS.

**Most important symptoms/effects, acute and delayed**: Fatal if inhaled. Causes serious eye irritation with tearing and redness. May cause respiratory irritation with coughing and sneezing.

**Indication of immediate medical attention and special treatment needed, if necessary**: Avoid contact with the product to help the victim. Keep victim warm and quiet. Symptomatic treatment should comprise mainly supportive measures such as correction of electrolyte disturbances, metabolic, and respiratory support. In case of skin contact do not rub the affected site.

## 5 – FIRE-FIGHTING MEASURES

**Suitable extinguishing media:** Suitable: Compatible with foam, carbon dioxide ($CO_2$), water mist and dry chemical powder.

Not recommended: Water jets directly.

**Specific hazards arising from the chemical**: The combustion of the chemical products or containers may form toxic and irritating gases such as carbon monoxide and carbon dioxide.

**Special protective actions for fire-fighters**: Self-contained breathing apparatus (SCBA) operated in positive pressure mode and complete protective clothing. Keep containers cool with water spray.

## 6 – ACCIDENTAL RELEASE MEASURES

**Personal precautions, protective equipment and emergency procedures**

**For non-emergency personnel:** Prevent sparks or flames. Do not smoke. Evacuate the area within a radius of at least 50 meters. Keep unauthorized persons away from the area. Stop the leak if this can be done without risk. Do not touch damaged containers or spilled material without wearing suitable clothing. Avoid exposure to the product. Use personal protective equipment as described in section 8.

**For emergency responders:** Use personal protective equipment with safety goggles with side protection, safety gloves, waterproof overalls and PVC boots. In case of leakage, where the exposure is high, it is recommended to wear a respiratory protection mask with activated carbon filter.**Environmental precautions:** Avoid spilled material reaches waterways and sewers.

**Methods and materials for containment and cleaning up**: Paved floor: absorb the product with sawdust or sand, collect the material with a shovel and place in sealed container and properly identified. The spilled product should not be used again. Soil: remove layers of

contaminated soil until reaching uncontaminated soil, collect this material and place in a properly labeled sealed container. Bodies of water: immediately stop the collection for human or animal consumption, contact the nearest environmental agency and the company's emergency center, since the measures to be adopted depend on the proportions of the accident, the characteristics of the water body in question And the quantity of product involved. For final destination, proceed according to Section 13 of this MSDS.

## 7 – HANDLING AND STORAGE

**Precautions for safe handling**

**Safe handling of the substance or mixture:** Handle in a well ventilated area or with general system of ventilation/local exhaust. Avoid vapors and mists formation. Avoid exposure to the chemical. Avoid contact with incompatible materials. Use personal protective equipment as described in Section 8.

**General hygiene**: Wash hands after use. Eating, drinking and smoking in work areas is prohibited. Remove contaminated clothing and protective equipment before entering eating areas.

**Conditions for safe storage, including any incompatibilities**:

**Technical measures for prevention of fire and explosion:** It is not expected that the product presents a fire or explosion hazard.

**Adequate conditions**: Store in a well ventilated place away from sunlight. Keep container closed. Keep stored at room temperature. It is not necessary the addition of stabilizers and antioxidants to ensure the durability of the product.

**Packaging compatibilities**: Plastic packaging designed and manufactured in such a way as to prevent any leakage, evaporation, loss or alteration of its contents and to facilitate the operations of washing, grading, reuse, recycling and proper disposal. Must be provided with seal, and immune to the action of its contents or insusceptible to form with it harmful or dangerous compounds.

## 8 – EXPOSURE CONTROLS, PERSONAL PROTECTION

**Control parameters**

**Occupational exposure limit**: Not established.

**Biological limit**: Not established.

**Appropriate engineering controls**: Promote direct mechanical ventilation and exhaust system to the outside environment. These measures help reduce exposure to product.

**Individual protection measures, such as personal protective equipment (PPE)**

**Eye/face protection**: Safety goggles with side protection.

**Skin protection**: Safety gloves, waterproof overalls and PVC boots. **Respiratory protection**: Respiratory protection mask with activated carbon filter **Thermal hazards**: Does not present thermal hazards.

## 9 – PHYSICAL AND CHEMICAL PROPERTIES

**Physical state:** Turbid homogeneous liquid (Concentrated Suspension).

**Colour:** White

**Odour:** Characteristic

**Melting point/freezing point:** Not available.

**Boiling point or initial boiling point and boiling range:** Not available.

**Flammability:** Not available.

**Lower and upper explosion limit /flammability limit:** Not available.

**Flash point:** Not available.

**Auto-ignition temperature:** Not available.

**Decomposition temperature:** Not available.

**pH:** 5.78 at 19.5 ° C (Pure sample) and 6.01 at 20 ° C (1% solution)

**Kinematic viscosity**: 1175 mPa.s at 40°C; 2378.3 mPa.s at 20°C

**Solubility**: There is no separation of liquid phase for water, acetone and ethanol.

**Partition coefficient n-octanol/water (log value)**: Not available.

**Vapour pressure**: Not available.

**Density and/or relative density:** 1.36 g/cm$^3$ at 19,6°C

**Relative vapour density**: Not available.

**Particle characteristics**: Not available.

**Other information**:

Corrosivity: 0.0115mm / year at 24.1°C after 7 days the substance.

Surface tension: 50.81x10$^{-3}$ N/m at 25°C

## 10 – STABILITY AND REACTIVITY

**Reactivity:** Reactivity is not to be expected under normal conditions of temperature and pressure**.**

**Chemical stability:** The product is stable under normal conditions of storage and use.

**Possibility of hazardous reactions:** There are not known hazardous reactions with the product.

**Conditions to avoid**: Elevated temperatures. Direct incidence of sunlight.

**Incompatibility:** There are not known incompatible materials with the product.

**Hazardous Decomposition Products:** Burning may produce toxic fumes.

## 11 – TOXICOLOGICAL INFORMATION

**Acute toxicity**: This product has been subjected to acute toxicity tests. Fatal if inhaled. Product not classified as acute toxic by oral and dermal route.

$LC_{50}$ (inhalation, vapors and mists, 4h, rats): 0.16mg/L

$LD_{50}$ (oral, rats): >2000 mg / kg

$LD_{50}$ (dermal, rats): >4000 mg / kg

**Skin corrosion/irritation**: This product has been subjected to skin irritation tests and has not been classified as an irritant.

**Serious eye damage/irritation:** Causes serious eye irritation with tearing and redness.

**Respiratory or skin sensitization:** This product has been subject to skin sensitization tests and has not been classified as a skin sensitizer. This product has not been subjected to respiratory sensitization tests and no studies on the ingredients are available.

**Germ cell mutagenicity:** This product was tested for germ cell mutagenicity and did not induce a mutagenic effect on the marrow of mice after oral administration at 24 hour intervals at the doses of 80, 160 and 320 mg/kg body weight.

Ames test: Negative.

**Carcinogenicity:** Suspected of causing cancer.

Information regarding to:

- Chlorothalonil:

Possibly carcinogenic to humans (Group 2B – IARC).

**Reproductive toxicity:** No data available.

**STOT - Single exposure:** May cause respiratory irritation with coughing and sneezing.

**STOT - Repeated exposure:** No data available.

**Aspiration hazard:** No data available.

## 12 – ECOLOGICAL INFORMATION

**Toxicity:** This product has been subjected to acute toxicity tests.

Very toxic to aquatic life with long lasting effects.

$EC_{50}$ (*Daphinea similis*, 48h): 0.15 mg/L

$LC_{50}$ (*Pseudokirchneriella subcapitata*, 72h): 0.19 mg/L

**Persistence and degradability:** It is expected that the product presents persistence and it is not considered readily biodegradable.

**Bioaccumulative potential:** No data available.

**Mobility in soil:** Low mobility in soil.

**Other adverse effects:** There are not known adverse environmental effects of the product.

## 13 – DISPOSAL CONSIDERATIONS

**Disposal methods:** Must be disposed of as hazardous waste in compliance with local regulations. The treatment and disposal should be evaluated for each specific product. Keep the product remains in its original and properly closed. Disposal should be performed as established for the product. Do not reuse empty containers. These may contain product residues and should be kept closed and sent for proper disposal as established for the product.

## **14 –** TRANSPORT INFORMATION

**Road:** UN - "United Nations"

Recommendations on the TRANSPORT OF DANGEROUS GOODS. Model Regulations

**UN Number:** 2902

**UN Proper Shipping Name:** PESTICIDE, LIQUID, TOXIC, N.O.S. (Chlorothalonil)

**Transport hazard class(es):** 6.1

**Packing group:** II

**Rail:** Convention concerning International Carriage by Rail (COTIF)

Appendix C - Regulations concerning the International Carriage of Dangerous Goods by Rail - RID

**UN Number:** 2902

**UN Proper Shipping Name:** PESTICIDE, LIQUID, TOXIC, N.O.S. (Chlorothalonil)

**Transport hazard class(es):** 6.1

**Packing group:** II

**Sea:** IMO – International Maritime Organization International

Maritime Dangerous Goods Code (IMDG Code) **UN**

**Number:** 2902

**UN Proper Shipping Name:** PESTICIDE, LIQUID, TOXIC, N.O.S. (Chlorothalonil)

**Transport hazard class(es):** 6.1

**Packing group:** II

**Marine pollutant:** The product is not considered a marine pollutant.

**EmS:** F-A, S-A

**Air:** IATA - International Air Transport Association

Dangerous Goods Regulation (DGR)

**UN Number:** 2902

**UN Proper Shipping Name:** PESTICIDE, LIQUID, TOXIC, N.O.S. (Chlorothalonil)

**Transport hazard class(es):** 6.1

**Packing group:** II

**Environmentally hazardous:**The product is considered environmentally hazardous.

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code:** Consult regulations:

- International Maritime Organization. MARPOL: Articles, protocols, annexes, unified interpretations of the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto, consolidated edition. IMO, London, 2006.

- International Maritime Organization. IBC code: International code for the construction and

equipment of shipping carrying dangerous chemicals in bulk: With Standards and guidelines relevant to the code. IMO, London, 2007.

## 15 – REGULATORY INFORMATION

**Chinese Regulations:**

Regulations on Pesticide Administration.

The Safety Management for Hazardous Chemical Statute.

**International Regulations:**

Convention Concerning Safety in the Use of Chemicals at Work (C170).

Guidelines for the Safe Warehousing of Pesticides.

Guidelines for the Safe Transport of Pesticides.

Convention concerning Safety in the use of Chemicals at Work (Convention 170) - International Labour Organization, 1990.

## 16 – OTHER INFORMATION

This SDS was prepared based on current knowledge about the proper product handling and under normal conditions of use, in accordance with the application specified on the packaging. Any other use of the product involving their combination with other materials, and use various forms of those indicated, are the responsibility of the user. Warns that the handling of any chemical substance requires the prior knowledge of its hazards for the user. In the workplace it is for the user company's product promotes training of its collaborators about the possible risks arising from exposure to the chemical.

SDS elaborated in April 2017.

**Change control:**

| Version | Publication date | Changes |
|---|---|---|
| 01 | 04/12/2017 | Translate |

Abbreviations:

**CAS –** Chemical Abstracts Service

**$LC_{50}$ –** Lethal Concentration 50%

**IARC –** *International Agency for Research on Cancer*

**Referências bibliográficas:**

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIALS HYGIENISTS. TLVs® and BEIs®: *Baseados na "Documentação" dos Limites de Exposição Ocupacional* (TLVs®) *para Substâncias Químicas e Agentes Físicos & Índices Biológicos de Exposição* (BEIs®). *Tradução Associação Brasileira de Higienistas Ocupacionais. São Paulo, 2016.*

Globally Harmonized System of Classification and Labelling of Chemicals (GHS). 6. rev. ed. New York: United Nations, 2015.

Pilarquim BR Comercial Ltda – FISPQ – *Ficha de Informações de Segurança de Produtos* Químicos – ICHIBAN – São Paulo/Brasil. Revision 01. March 2017